Mitigate the LFM-PRFI in SAR Data: Joint Down-Range and Cross-Range Filtering

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***Abstract*— Synthetic aperture radar (SAR), an active remote sensing equipment, shares the spectrum with devices in the same frequency band and is, therefore, easy to affect by pulse radio frequency interference (PRFI). The wideband version of PRFI, i.e., linear-frequency-modulation PRFI (LFM-PRFI), derived from the ground- and space-based radar sensors is a challenging issue for SAR because it usually has a large bandwidth and pulsewidth compared with the traditional PRFI. The well-known notch filtering methods, including time- and frequency-domain versions, are robust approaches against PRFI, which has been integrated into some SAR ground processing systems. However, the down-range frequency-domain notch filtering will increase the sidelobes of the targets when mitigating wideband LFM-PRFI, whereas the time-domain version will introduce ghosts when notching large pulsewidth ones. This article proposed a filtering method that consists of three steps to tackle the above problems. The first step is focusing the energy of LFM-PRFI down-range and cross-range simultaneously, which can be done by match filtering and Fourier transform. The second step is mitigating the LFM-PRFI, which can be done by a designed filter. The last step is data restoration, which can be done by multiplying the conjugate of the phase function used before. Numerical experiments based on simulated SAR data and measured spaceborne SAR data acquired by European Sentinel-1 are performed to test the mitigation performance, which verified the effectiveness and superiority of the proposed approach.**

***Index Terms*— Notch filtering, pulse radio frequency interference (PRFI), synthetic aperture radar (SAR).**

I. Introduction

A. Background

THE concept of synthetic aperture radar (SAR) is proposed by Wiley et al. in the early 1950s, and it breaks through the limitation of traditional radar. SAR is an active imaging sensor that emits wideband signals in range and obtains high resolution of the targets with the help of pulse compression technology. It achieves high resolution by employing synthetic aperture technology in azimuth. SAR can provide day-and-night, weather-independent, and high-resolution images compared to optical imaging sensors, and it has been widely used for remote sensing for a multitude of applications ranging from military reconnaissance, resource exploration, topographic mapping, environmental monitoring, disaster warning and assessment, and other related fields [1], [2], [3]. After decades of development, rich operation modes, such as stripmap, scan, terrain observation by progressive scans (TOPS), spotlight and sliding spotlight, and advanced concepts, such as digital beamforming, multiple-input multiple-output, and tomography, are being explored to meet increasing user requirements [4], [5], [6].

Modern communication technology has entered a golden age. Systems such as mobile, radio, TV, and cellular networks are active, and therefore, the useable spectrum gets increasingly crowded [7]. In a complex electromagnetic environment, SAR is an open radar system, and it shares the spectrum with equipment in the same frequency band, which means that the SAR is easy to receive radio frequency interference (RFI) signals [8]. The pulsed version of RFI, i.e., pulse RFI (PRFI), has serious impacts on the SAR system specifically affecting signal processing and image interpretation, thereby directly hindering the subsequent applications. PRFI belongs to a one-way direct wave signal with high energy and a certain pulse repetition frequency (PRF) compared with the SAR signal in the form of a two-way scattered wave. PRFI artifacts behave differently on SAR products under different stages.

1) Raw Product: PRFI artifacts appear as pseudoperiodic bright stripes down-range on the SAR raw data.

2) Single Look Complex (SLC) and Above Level Image Product: PRFI is forced to add 2-D phase modulation through the SAR imaging processing and defocused in the SAR image and specifically behaves like periodic bright stripes, raindrops, fog, and so on since it mismatched with the matched filter of the SAR. Two typical cases are given in Fig. 1 to illustrate artifacts in PRFI-contaminated products, where the artifacts differ at various stages of the SAR product.

Researchers generally consider that the PRFI phenomenon is serious in low-frequency (such as P- and L-bands) SAR systems, of which the Japanese Phased-Array L-Band SAR (PALSAR) system is a typical case. The flagship satellite,

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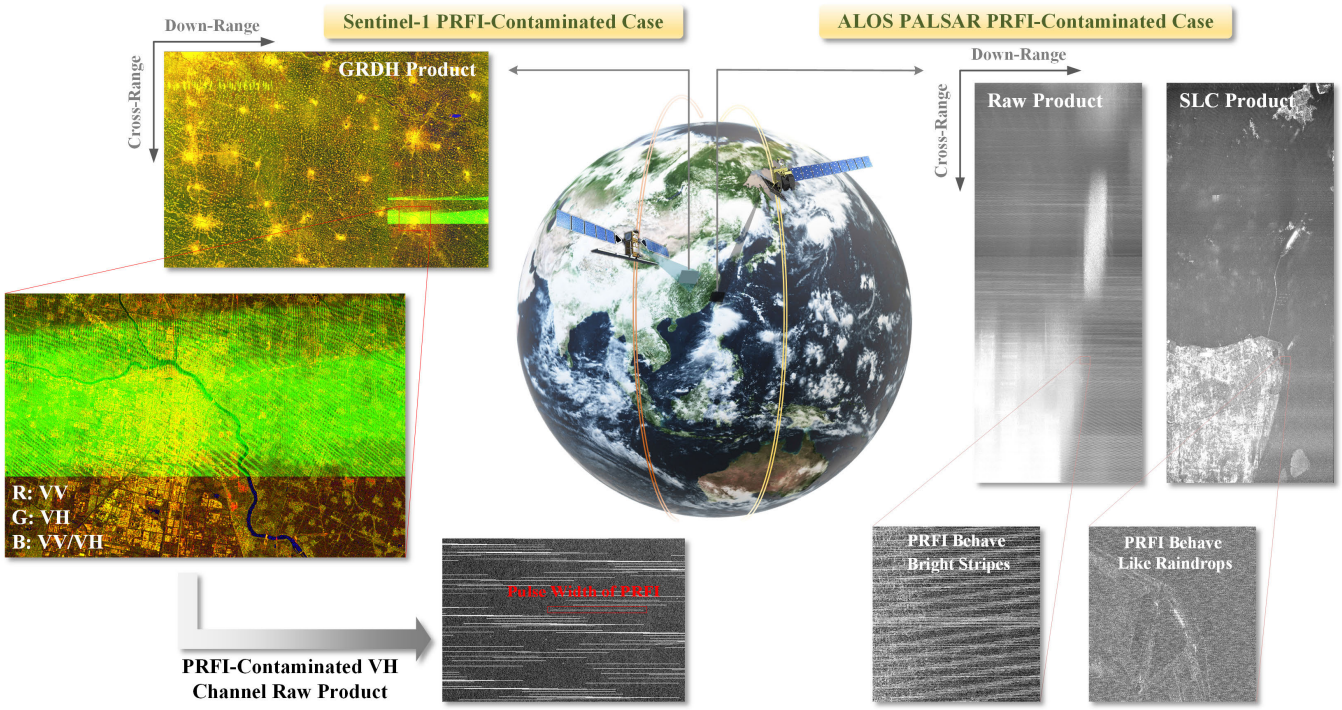


Fig. 1. Cases of PRFI-contaminated SAR products.

namely, the Advanced Land Observing Satellite (ALOS) designed by the Japan Aerospace Exploration Agency, was launched on January 24, 2006. The core sensor on ALOS, i.e., the PALSAR system, was expected to perform specific Earth observation tasks by virtue of its excellent penetration capabilities compared to C- and X-band SARs [9]. However, the PRFI in the L-band seriously affects the radiometric correction, interferometry, and other applications of PALSAR. The phenomenon of long-duration PRFI-related image quality degradation was often been observed, and many fixed areas were often contaminated. Interference detection and mitigation-related modules were integrated into PALSAR's ground processing systems to solve the problems. In fact, RADARSAT-1, which was launched on November 4, 1995, by the Canadian Space Agency (CSA) and equipped with a C-band SAR sensor (carrier frequency is 5.400 GHz), has been found to be frequently interfered with by ground communication systems (working on 5.400 GHz). Later, CSA adjusted the carrier frequency to 5.405 GHz when designing RADARSAT-2 to avoid the above interference. PRFI-contaminated cases are increasingly common in C-band SAR sensors. The European Space Agency (ESA) Sentinel-1 mission was not immune to the effects of the interference. The Sentinel-1 Mission Performance Centre (MPC-S1) publicly reported RFI issues in Sentinel-1 in 2015 [10], 2018 [11], and 2022 [12], respectively. In particular, Sentinel-1 has had a phenomenon of mutual RFI (MRFI) between satellites for a long time. The MRFI signals come from C-band SAR sensors, including Canadian RADARSAT-2 and RCM1/2/3, and Chinese Gaofen-3 [13], [14], [15], where the MRFIs are mainly scattered linear-frequency-modulation (LFM) signals with large bandwidth. Aresys and ESA provide viewers with information (maps and classification) about the interfered cases as observed from C-band measurements by Sentinel-1 [16]. In early 2022, ESA integrated interference detection and suppression technology into the Sentinel-1 Instrument Processing Facility (S1-IPF), where the detailed processing was recorded in [12]. The problem of interference is gradually being taken seriously, whereas the form of PRFI has expanded from narrowband signals to wideband ones. Especially, the common form of wideband PRFI, i.e., LFM-PRFI derived from the ground- and space-based radar sensors, is a challenging issue because its power, bandwidth, and pulsewidth are higher than that of the traditional ones. Many of the above cases show that the phenomenon of PRFI is intensifying, and therefore, the research on mitigation means is of great significance to improving the survivability and efficiency of SAR sensors in complex electromagnetic environments.

B. State of the Art of RFI Mitigation

Researchers have studied the properties of unintentional RFI for several years to find effective and robust removal approaches. Advanced techniques, such as spectral estimation [17], sparse representation [18], compressed sensing [19], matrix factorization [20], signal separation [21], and machine learning [22], are successfully applied to mitigate RFI. Typically, these methods can be divided into three categories, namely, parametric methods, nonparametric methods, and semiparametric methods [23].

1) Parametric Methods: Early studies found that RFI in the SAR spectrum is usually distributed at fixed frequency with small bandwidth. Researchers modeled this kind of RFI, namely, narrowband interference (NBI), as the superposition of sinusoidal signals, and employ approaches such as maximum

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likelihood estimation [24], least-squares estimation [25], and least mean square [26] and its modifications [27] to eliminate NBI [28]. Researchers in the U.S. Army Research Laboratory (ARL) carried out a lot of demonstration studies [29]. A novel system [30], which can collect RFI-only data, is designed to boost the estimation accuracy of the parameter. The well-known theory of modern spectral estimation, i.e., the iterative adaptive approach (IAA), is employed to estimate the parameters of NBI, whereas its modification is extended to wideband interference (WBI) mitigation [17].

In measured SAR data, RFI is not only in the narrowband form, and it is increasingly common in the form of wideband version as aforementioned. WBI is generally modeled as frequency-modulated (FM) signals, such as LFM and sinusoidal FM (SFM) signals. Zhou et al. have carried out many studies on WBI mitigation, including regularized time–frequency filtering [31] and second-order multisynchrosqueezing transform [32].

2) Nonparametric Methods: Nonparametric methods generally establish the feature map of the RFI in the transform domains and mitigate it through the discrimination criterion (usually based on the energy distribution). Notch filtering approaches, including time [33], [34], frequency [28], [35], and time–frequency-domain [21] versions and their modification, are best known. The robust frequency-domain notch filtering (FNF) method has been widely performed in engineering. The German Airborne E-SAR and Japanese PALSAR systems have integrated the FNF method into their ground data processing ground station. In early 2022, ESA integrated FNF and time-domain notch filtering methods into S1-IPF. Technologies such as the linear prediction model [36], sparse representation, compressed sensing [19], and others have been studied to recover the missed spectrum gap, which avoids the abnormal target response produced by the FNF method when against WBI; Li et al. employed missing data IAA (MIAA) [33] and low-rank matrix completion [34] techniques to recover the missed gaps in raw data, which avoids the ghosting introduced by the periodic missing data of the TNF method. With the prior that the subspace of the signal RFI is orthogonal, Zhou et al. [37] proposed the eigen-subspace projection (ESP) method. In addition, methods based on independent component analysis [38], complex empirical mode decomposition [39], and independent space analysis [40] are proposed by Zhou et al. [37], which gets well mitigation performance.

In practice, it is difficult for users to obtain SAR raw product, which makes the above methods unusable, and therefore, image domain methods, such as subband spectral cancellation [41] and its modifications [42], principal component analysis and its robust version, block subspace filter [43], 2-D spectral analysis filter [44], and image-domain sparse regularization [45], are proposed. Most of these methods work on publicly available SLC products and are able to mitigate specific types of RFI.

3) Semiparametric Methods: Sparse signal processing and low-rank decomposition theory have been successfully applied in the field of RFI mitigation in recent years. This type of method converts the signal separation problem into a hyperparameter optimization problem and forms the semiparametric method. Nguyen et al. develop sparse recovery-based studies to mitigate RFI [46] and upgrade the approach to joint down-range and cross-range versions in subsequent research [47], which improved the mitigation performance. A series of "low-rank and sparse" models [49], [50], [51], [52] is developed by Huang et al., where the mitigation capability in the model is realized by restraining RFI, and the data protection capability is realized by restraining SAR signal. Recent research has focused on conservation constraints on SAR signals and sparse constraints on RFI in the model. Besides, some nonconvex models [53] have also been developed.

It should be noted that operations such as iterative processing and matrix inversion inevitably introduce a huge amount of calculation in performing semiparametric methods, and therefore, it is more suitable for off-line processing than the other ones.

C. Problem Statement

In general, narrowband PRFI can be well mitigated by the FNF method, and the missed fixed frequency gaps hardly affect the subsequent processing of SAR data. It is widely known that FNF is powerless in dealing with wideband PRFI. The TNF is proposed as a solution, thanks to it being performed in the time domain, and it is insensitive to the bandwidth of the PRFI. However, the LFM-PRFIs have long been a thorny issue because it has a larger bandwidth and pulsewidth and, in severe cases, overlap with the entire SAR spectrum [14]. In particular, the emerging LFM-PRFI in recent years, i.e., the mutual interference in spaceborne SAR (the interference comes from the SAR sensor in the same frequency band), causes serious deterioration to the SAR data of interest [44]. The mitigation performance of TNF and other types of methods will be greatly reduced against LFM-PRFI.

D. Solution and Contributions

This article focuses on mitigating the particular PRFI in SAR data, i.e., LFM-PRFI, and proposes a joint down-range and cross-range filtering method that overcomes the limitations of traditional FNF and TNF methods. The proposal consists of three steps.

1) The first step is focusing on LFM-PRFI both in down-range and cross-range, wherein the FM rate of the LFM-PRFI is estimated by the processing of cross correlation. Then, the LFM-PRFI can be focused down-range by the estimated FM-rate and can be focused in cross-range by shifting them in a fixed reference range and Fourier transform.

2) The second step is filtering the focused LFM-PRFI, wherein a mask that is performed in a special domain is designed.

3) The third step is data restoration, wherein it can be achieved through phase compensation.

In comparison to the existing works, the main contributions of this article are summarized as follows.

1) The model, mechanism, and characteristics of LFM-PRFI are introduced for the first time in this article.

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Multiple real cases of LFM-PRFI-contaminated data acquired from ESA Sentinel-1 are shown and analyzed.
2) A joint down-range and cross-range filtering for LFM-PRFI mitigation is proposed. The proposal for the first time focuses the energy of LFM-PRFIs in two directions and then filters them out. In contrast to the traditional notch filter-based method, the proposal does not cause a series of abnormal target responses, such as azimuth ghosts and high-range sidelobes.
3) Extensive experiments using simulated and measured SAR data are performed. Specifically, experiments based on simulation SAR data are employed to analyze the mitigation performance of the proposed approach, whereas experiments based on real LFM-PRFI-contaminated data acquired from ESA Sentinel-1 are employed to verify the effectiveness and superiority of the proposed method.

E. Organization

The remainder of this article is organized as follows. In Section II, the knowledge of PRFI including geometry, signal models, and characteristics is introduced. In Section III, the proposed joint down-range and cross-range filtering approach for removing LFM-PRFI is presented in detail. In Section IV, extensive experimental results, qualitative and quantitative assessment analysis, and a discussion of the proposal are given. Section V briefly concludes this article.

II. Knowledge of PRFI: Geometry, Signal Models, and Characteristics

A. Geometric Relationship Between SAR and Interferer

The SAR system works in the microwave band and is an open system that actively transmits pulse signals to the region of interest (ROI). Its receiver actively receives the ground echo, and at the same time, the unwanted signals generated by equipment in the same frequency band will be unintentionally received by SAR. These unwanted signals must meet certain restrictions before they may interfere with SAR, which are listed as follows.

1) The first is spatial overlap, which means that the beam footprint of SAR needs to cover the interferer. At the same time, the working beam of the interferer needs to illuminate SAR.
2) The second is time overlap, which means that the working time of the SAR and the interferer overlaps, and the arrival times of the RFI need to be within the receiving duration of the receiver.
3) The third is frequency overlap, which means that the interferer overlaps or even covers the working frequency of the SAR.
4) The last is the overwhelming power of the interferer. SAR emits a chirp signal with a large time-bandwidth product to sense the ROI. It can achieve a gain of 60 dB or even higher through the coherent accumulation of pulses. The power of the RFI needs to be much higher than that of the SAR signal under the circumstances.

An explanatory geometric image is shown in Fig. 2 for easy understanding. Fig. 2(a) shows the geometric relationship between SAR and interferer, where an interferer is placed in the center of the scene, and it transmits PRFI to SAR pulse-by-pulse. $P_{\mathrm{IC}}(X_c, Y_c, 0)$ records the position of the interferer. An airborne SAR platform flies at a height H at a speed of V_r, and $P_{\mathrm{SAR}}(0, Y_v, H)$ records the position in the 3-D coordinate system. R_0 and R denote the slant range and instantaneous range between the interferer and the nominal track, respectively. α and $\theta_{r,c}$ denote the look angle and squint angle, respectively. Fig. 2(b) shows a simulated scenario with 5-km swath width, where five targets are placed with an interval of 1.6 km in range and 1 km in azimuth. Fig. 2(c) shows the SAR transmission and reception timing diagram, where $\mathrm{PRT}_{\mathrm{PRFI}}$ and $\mathrm{PRT}_{\mathrm{SAR}}$ denote the pulse repetition time of SAR and interferer, respectively. T_r and T_d denote the pulsewidth and receiving duration of the SAR, respectively. PRFI enters the SAR receiver when the arrival time of PRFI overlaps with the receiving duration of the SAR. Without loss of generality, the following analysis is based on the above geometric relationship.

B. Signal Models of PRFI

Generally, the data collected by the SAR receiver are in 2-D form, and the time along the range is defined as the fast time, whereas the ones that cross the range direction are defined as the slow time. After the receiver undergoes quadrature demodulation and digital sampling, the received data containing PRFI can be expressed as

$$s(\tau, \eta) = r(\tau, \eta) + n(\tau, \eta) + i(\tau, \eta) \tag{1}$$

where τ and η denote the fast (down-range) and slow (cross-range) times, respectively. $r(\tau, \eta)$, $n(\tau, \eta)$, and $i(\tau, \eta)$ denote the two-way backscattered echo, system noise, and PRFI, respectively. The analytical expression of backscattered echo $r(\tau, \eta)$ is not discussed here, and system noise $n(\tau, \eta)$ will not be considered in the following.

PRFI belongs to a one-way direct wave signal with high energy and a certain PRF, which is mainly derived from ground-based radar equipment. The signal model of PRFI can be expressed as

$$\begin{aligned} i(\tau, \eta) = \sum_{n=1}^{N} A_n \mathrm{rect}\left(\frac{\tau}{T_d}\right) \mathrm{rect}\left(\frac{\tau - \tau_d(\eta)}{T_n}\right) \\ \times \exp(j(2\pi f_n(\tau - \tau_d(\eta)) + \varphi_n(\tau - \tau_d(\eta)))) \end{aligned} \tag{2}$$

where N is the number of PRFI components and $n = 1, 2, \ldots, N$; A_n and f_n denote the amplitude modulation and the working frequency; T_d denotes the receiving duration of the SAR, which can be calculated by

$$T_d = \frac{2(R_f - R_n)}{c} + T_i \tag{3}$$

where R_f, R_n, c, and T_i denote the farthest slant range, the nearest slant range, the velocity of light, and PRFI's pulsewidth, respectively. τ_d denotes the one-way delay of the

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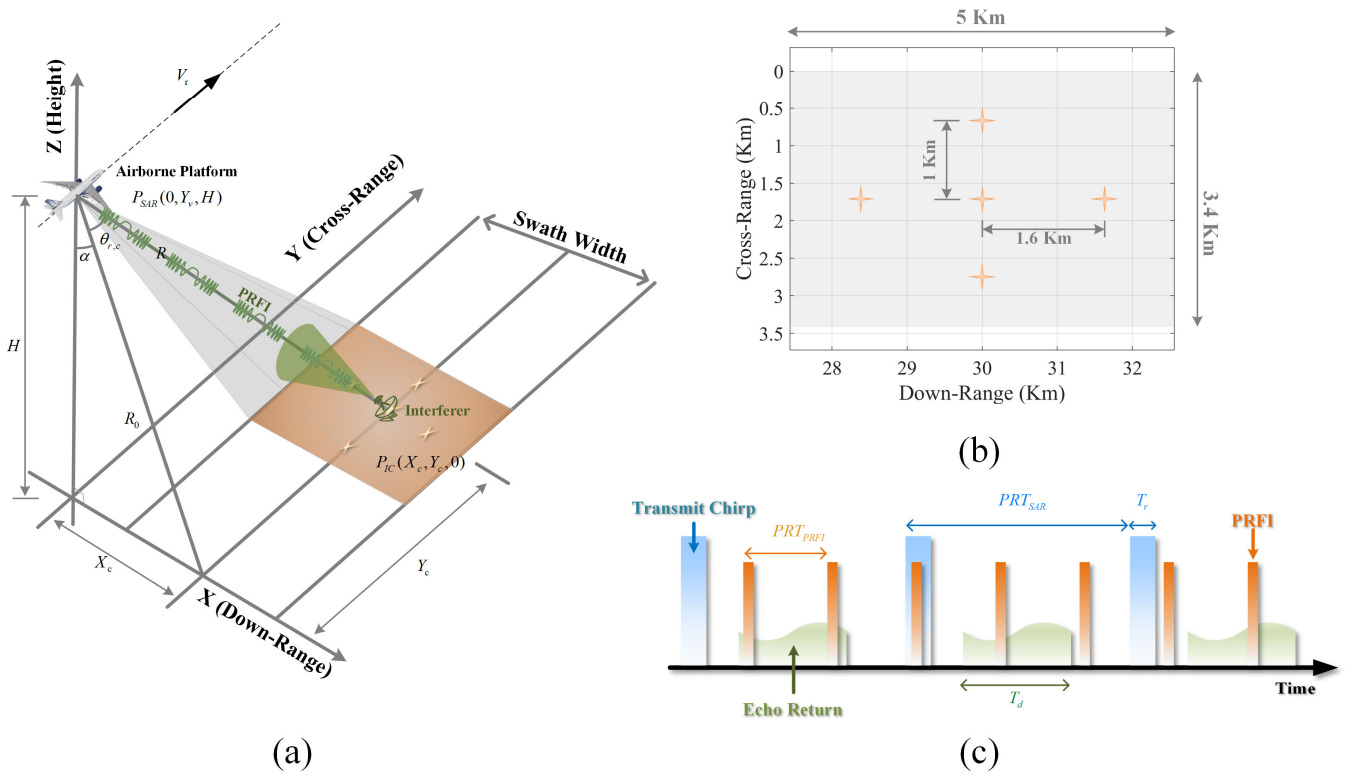


Fig. 2. Geometric relationship and transmission and reception timing diagram between SAR and interferer. (a) Geometric relationship between SAR and interferer. (b) Simulated scenario with 5-km swath width. (c) SAR transmission and reception timing diagram.

PRFI, which can be calculated by

$$\tau_d(\eta) = \frac{|P_{\mathrm{SAR}} P_{\mathrm{IC}}|}{c} = \frac{\sqrt{X_c^2 + (Y_c - Y_v)^2 + H^2}}{c} = \frac{\sqrt{X_c^2 + (Y_c - V_r\eta)^2 + H^2}}{c}. \tag{4}$$

The type of PRFI depends on the phase term $\varphi_n(\tau)$. Specifically, $\varphi_n(\tau) = 0$ corresponding to the narrowband version of PRFI, whereas wideband versions are generally modeled as LFM and SFM signals, which can be expressed as

$$\varphi_{\mathrm{LFM}}(\tau) = \pi K_n \tau^2 \tag{5}$$

$$\varphi_{\mathrm{SFM}}(\tau) = \beta_n \sin(2\pi f_n \tau) \tag{6}$$

where K_n and β_n denote the FM frequency and factor in the nth component of PRFI, respectively. It should be noted that, when $\varphi_n(\tau)$ in formula (2) is $\varphi_{\mathrm{SFM}}(\tau)$, $2\pi f_n \tau$ in the exponent term in formula (2) should be ignored.

The LFM version of PRFI is very common in measured SAR data compared with the SFM ones. LFM waveforms are often used in radar sensors, and according to related studies [13], [14], [15], [44] and reports [11], ground- and space-based radars and even spaceborne SARs in the same frequency band may generate PRFI.

C. Characteristics of PRFI

The first characteristic of PRFI is predictably powerful, which is because it belongs to a one-way direct wave signal compared with the SAR signal in the form of a two-way scattered wave. The second characteristic of PRFI is time-varying, which is because it mainly originates from radar sensors that emit signals pulse-by-pulse. The simulated and measured SAR data are employed to characterize PRFI, among which three types of PRFI are simulated according to the scene in Fig. 2(b), and the characteristics of LFM-PRFI are analyzed based on the measured data of ESA Sentinel-1A. The simulated data aim to illustrate the frequency-modulation characteristics of different PRFIs, while the measured data aim to objectively analyze the time-varying and powerful features of PRFI.

TABLE I
MAIN PARAMETERS OF SIMULATION

Parameters	Values
Carrier Frequency	9.400 GHz
Sampling Frequency	120 MHz
Bandwidth	100 MHz
Efficient Velocity	250 m/s
Slant Range	30 Km
PRF	600 Hz
Pulse Width	10 μs
Carrier Frequency of PRFI	9.420 GHz
Pulse Width of PRFI	10 μs
PRF of PRFI	2827 Hz
SINR	-10 dB

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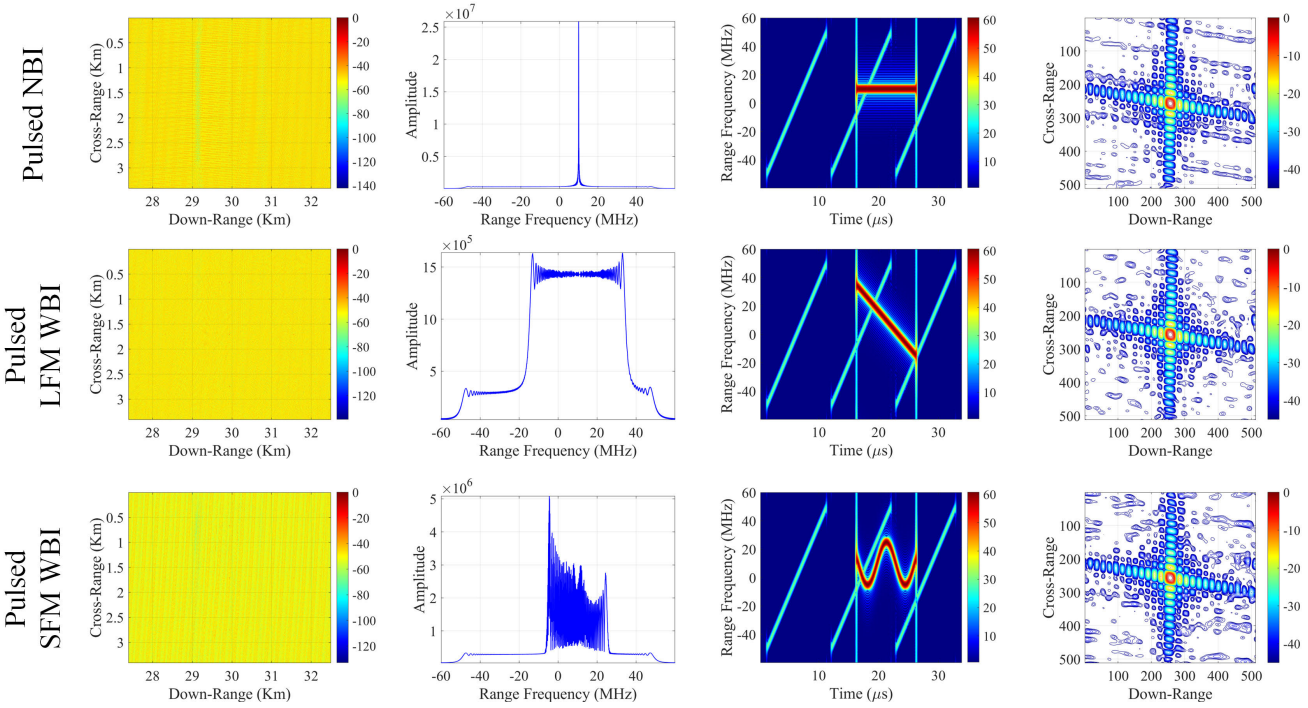


Fig. 3. Simulated PRFI in different domains. From (top) to (bottom) are pulse NBI, LFM WBI, and SFM WBI, respectively. From (left) to (right) are imaging results, magnitude spectrum of the range profiles, time–frequency diagrams, and contour plots of the impulse response function for the center target.

1) On the Characteristics of PRFI in Simulated SAR Data: The main system parameters of the simulation are listed in Table I. The simulated PRFI in different domains are shown in Fig. 3. The interferer is set in the center of the scene according to the geometric relationship in Fig. 2(a). From top to bottom in Fig. 3 are pulse NBI, LFM WBI, and SFM WBI, respectively, whereas from left to right are imaging results, range spectrum slices, time–frequency diagrams, and contour plots of the impulse response function for the center target.

Five targets are arranged in the scene according to Fig. 2(b), whereas the target is indistinguishable after adding −10-dB PRFI according to Fig. 3. In addition, pulsed NBI occupies a small portion of the spectrum of the SAR and appears as a spike whose frequency does not change with time. In contrast, pulsed LFM and SFM occupy a large proportion of the spectrum, and the time–frequency diagrams show their frequency trends over time. The impulse response function of the five targets is seriously distorted after adding −10 dB to three types of PRFI.

2) On the Characteristics of PRFI in Measured SAR Data: The measured PRFI-contaminated SAR data are acquired by the ESA Sentinel-1 mission, which is shown in Fig. 4. Inspired by the global C-band RFI monitoring system (available at https://s1rfimap.aresys.it) [16], many interference-contaminated cases can be observed, and ESA distributes Sentinel-1 products for free. The illuminated area shown in Fig. 4(a) is located in Bozhou City, Anhui Province, China, which is the junction of Henan and Anhui Provinces. The working mode of data is an interferometric wide (IW) swath using the TOPSAR imaging technique, and the analyzed area is located in the fourth burst of the third swath, where the data ID is marked in Fig. 4(a).

Fig. 4(b) shows the Raw Level-0 data, where the PRFI is densely distributed on the SAR echo. Fig. 4(c) shows the range spectrum of the data, where the strong artifacts marked by the red boxes are frequency components of PRFI. It is worth noting that the available SAR bandwidth is 42.79 MHz, whereas the bandwidth of PRFI is 4.81 MHz and is accompanied by frequency variation. The S1-IPF mainly employs the notch filtering method to mitigate interference, which is inappropriate to deal with such a wideband case. Fig. 4(d) shows the imaging results of the raw data, where the significant artifacts of PRFI seriously covered the observation scene. The reason for this is that PRFIs are forced to add 2-D phase modulation by the SAR imaging processing and defocused on the SAR image. The defocused pulses are superimposed on each other and accumulate energy, thereby suppressing the SAR signal. Furthermore, one of the SAR pulses that interfered with two PRFI is shown in Fig. 4(e), where the step part marked by the red box is PRFI, and the signal-to-interference plus noise ratio (SINR) is −7.2 dB. The magnitude spectrum of the range profiles and the time–frequency spectrum are shown in Fig. 4(f) and (g), respectively. The PRFI in this case is in the form of an LFM wave and is time-varying, and its frequency components are superimposed to occupy 4 MHz in the SAR spectrum. Mitigating the LFM-PRFI, in this case, is undoubtedly challenging because it has features of large bandwidth, wide distribution, and both time- and frequency-variant.

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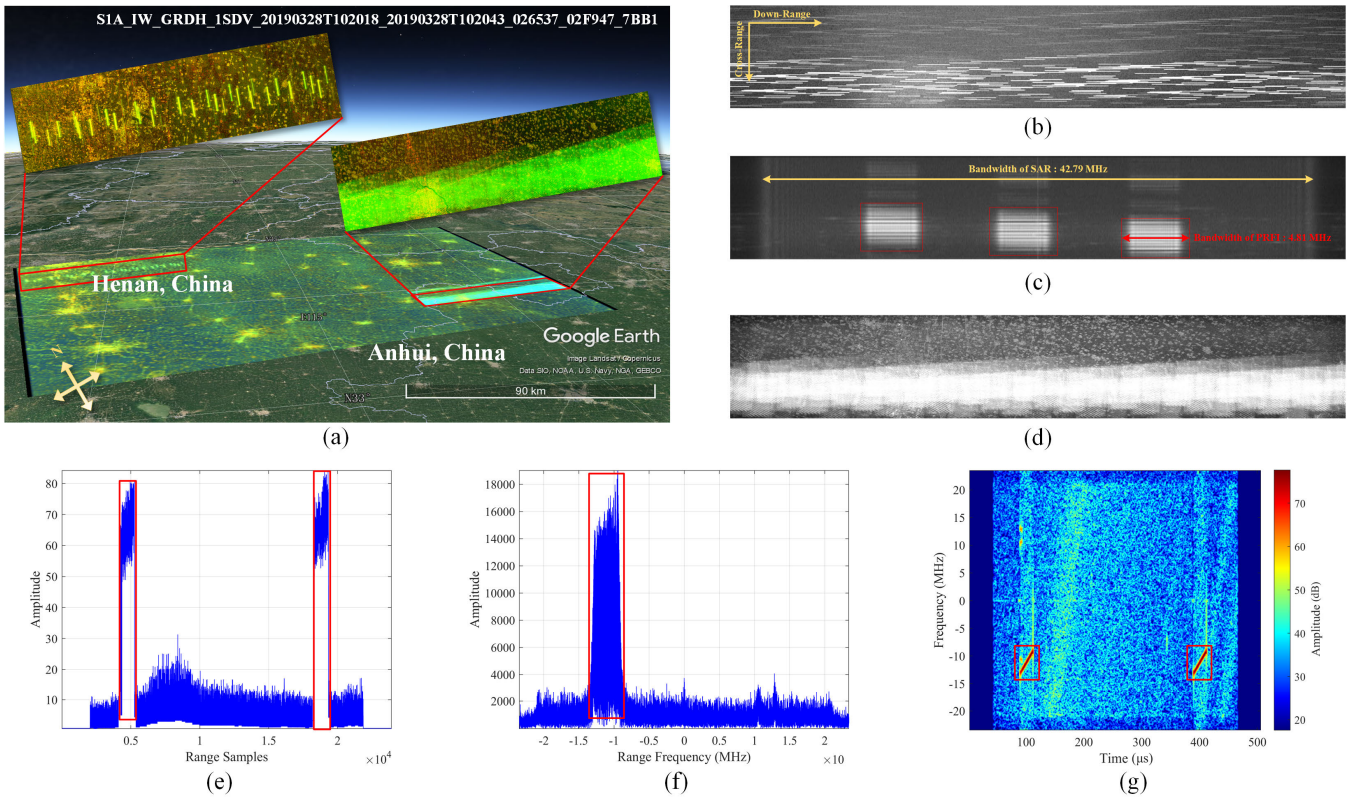


Fig. 4. Example case of the measured PRFI-contaminated SAR data. (a) Data illuminated area. (b) Amplitude image of raw data. (c) Amplitude image of the range spectrum. (d) Amplitude image of the ROI. (e) Amplitude of PRFI-corrupted pulses. (f) Magnitude spectrum of the range profiles. (g) Time–frequency diagram of (e).

The model, mechanism, and characteristics of PRFI in this section form the basis of the approach and experiments in this article.

III. Methodology: Joint Down-Range and Cross-Range Filtering for LFM-PRFI Mitigation

The bandwidth and pulsewidth of LFM-PRFI are generally higher than traditional PRFI, which means that the traditional notch filtering-based methods are not suitable for mitigating them. The missed spectral gaps caused by the FNF method will increase the target sidelobes, whereas the missed data caused by the TNF-based methods are difficult to recover, and seriously, ghosts and ambiguities will be introduced in the SAR imagery. A filtering method consisting of three steps, that is, joint down-range and cross-range processing, is proposed to tackle the above problems: 1) LFM-PRFI focusing; 2) LFM-PRFI removal; and 3) data restoration. The flowchart of the proposed approach is shown in Fig. 5.

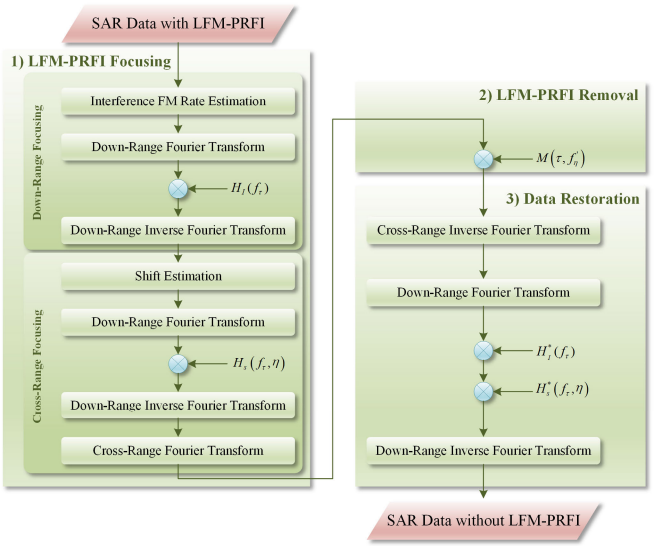


Fig. 5. Flowchart of the proposed approach.

A. LFM-PRFI Focusing

Modeling received SAR data with interference as (1) and ignoring the influence of system noise, expressed as

$$s(\tau,\eta) = r(\tau,\eta) + i(\tau,\eta). \tag{7}$$

The main purpose of interference mitigation is to remove $i(\tau,\eta)$ from $s(\tau,\eta)$ and obtain pure $r(\tau,\eta)$. The first step of the proposed method aims to focus the energy of LFM-PRFI, which facilitates the subsequent filtering process. The LFM-PRFI can be focused by matched filtering with a known FM rate. However, these dense PRFIs are still scattered in the SAR pulse, which is not conducive to notch filtering. Cross-range focusing is taken into account. The proposal shifts the focused PRFI to a fixed range cell pulse-by-pulse and treats it as a rectangular function, which can be focused by Fourier transform.

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1) Down-Range Focusing: As we all know, under the assumption of a known FM rate, the chirp signal can be focused into a sinc response function through matched filtering, and therefore, estimating the FM rate of LFM-PRFI is indispensable. LFM-PRFI generally has an obvious step component in the range frequency domain, and its frequency component is extracted by a threshold, which is empirically set to be the mean value plus the standard deviation of the spectrum amplitude.

Denoted the extracted spectrum of LFM-PRFI PRFI by $I_{\mathrm{LFM}}(f_\tau, \eta)$, its bandwidth can be expressed as

$$B_i = \arg\max_{f_\tau}\left[I_{\mathrm{LFM}}(f_\tau, \eta)\right] - \arg\min_{f_\tau}\left[I_{\mathrm{LFM}}(f_\tau, \eta)\right]. \quad (8)$$

Another parameter that needs to be estimated in order to estimate the FM rate is the pulsewidth of the LFM-PRFI, which can be achieved by cross correlation. The obtained spectrum of LFM-PRFI is divided into two subbands, expressed as

$$I_{\mathrm{LFM}_1}(f_\tau) = \mathrm{rect}\left(\frac{f_\tau - \frac{K_i T_i}{4}}{\frac{K_i T_i}{2}}\right)\exp\left(-j\pi\frac{f_\tau^2}{K_i}\right) \quad (9)$$

and

$$I_{\mathrm{LFM}_2}(f_\tau) = \mathrm{rect}\left(\frac{f_\tau + \frac{K_i T_i}{4}}{\frac{K_i T_i}{2}}\right)\exp\left(-j\pi\frac{f_\tau^2}{K_i}\right) \quad (10)$$

where K_i denotes the unknown FM rate of the LFM-PRFI and $B_i = |K_i T_i|$. f_τ denotes the range frequency. Two subsignals can be obtained by performing inverse Fourier transform on (9) and (10), where each of which represents half of the waveform of the original signal, and their time interval is $T_i/2$ delays. The analytical form of the two subsignals can be obtained by means of the principle of stationary phase (POSP), which can be expressed as

$$i_{\mathrm{LFM}_1}(\tau) = \mathrm{rect}\left(\frac{\tau - \frac{T_i}{4}}{\frac{T_i}{2}}\right)\exp\left(j\pi K_i \tau^2\right) \quad (11)$$

and

$$i_{\mathrm{LFM}_2}(\tau) = \mathrm{rect}\left(\frac{\tau + \frac{T_i}{4}}{\frac{T_i}{2}}\right)\exp\left(j\pi K_i \tau^2\right). \quad (12)$$

Carrying out cross correlation calculation on subsignals (11) and (12), the correlation between them is the highest at the delay $T_i/2$, and a peak will appear. Then, the estimated pulsewidth of PRFI can be expressed as

$$\begin{aligned} T_i &= 2 \times \arg\max_\tau\left[i_{\mathrm{LFM}_1}(\tau) \otimes i^*_{\mathrm{LFM}_2}(-\tau)\right] \\ &= 2 \times \arg\max_\tau\left[\int_{-\infty}^{\infty} i_{\mathrm{LFM}_1}(u) i^*_{\mathrm{LFM}_2}(u-\tau)du\right]. \end{aligned} \quad (13)$$

The designed matched filter for down-range focusing LFM-PRFI can be expressed as

$$H_I(f_\tau) = \exp\left(j\pi\frac{f_\tau^2}{K_i}\right). \quad (14)$$

Denoted the Fourier transform and its inverse version by $\mathcal{F}$ and $\mathcal{F}^{-1}$, the down-range focusing can be expressed as

$$\begin{aligned} &\tilde{s}(\tau, \eta) \\ &= \mathcal{F}_d^{-1}\left[\mathcal{F}_d[r(\tau, \eta) + i(\tau, \eta)] \times H_I(f_\tau)\right] \\ &= \mathcal{F}_d^{-1}\left[\left[R(f_\tau, \eta) + I(f_\tau, \eta)\right] \times H_I(f_\tau)\right] \\ &= \mathcal{F}_d^{-1}\left[R(f_\tau, \eta) \times H_I(f_\tau)\right] + \mathcal{F}_d^{-1}\left[I(f_\tau, \eta) \times H_I(f_\tau)\right] \\ &= \tilde{r}(\tau, \eta) + B_i \mathrm{sinc}(B_i \tau) \end{aligned} \quad (15)$$

where $\tilde{r}(\tau, \eta)$ denotes the SAR echo signal that is added with phase modulation $H_I(f_\tau)$, whereas $B_i\mathrm{sinc}(B_i\tau)$ denotes the focused LFM-PRFI. The subscript d of $\mathcal{F}$ indicates down-range processing.

2) Cross-Range Focusing: The PRFIs are time-varying aforementioned in Section II, which means that their positions in each SAR pulse are different after down-range focusing. Within the proposal, the concept of focusing PRFI at the cross-range to further gather the energy of PRFI is proposed. Inspired by the fact that PRFIs can be approximated as a rectangular function in cross-range if they are shifted to a fixed reference range, therefore, it is easy to focus the PRFIs in the frequency domain along the cross-range direction according to the nature of the Fourier transform.

The phase carrying time-varying shifts is designed, which can be expressed as

$$H_s(f_\tau, \eta) = \exp(2j\pi f_\tau T_s(\eta)) \quad (16)$$

where T_s denotes the time offset from the reference range, and it can be calculated by

$$T_s(\eta) = \arg\max_\tau\left[\left|\tilde{s}(\tau, \eta)\right|\right] - \frac{2R_{\mathrm{ref}}}{c} \quad (17)$$

where R_{ref} is the reference range. The setting of R_{ref} is arbitrary, and the author empirically sets it at the center of the scene. The focused PRFIs can be shifted to a reference range by appending a linear phase $H_s(f_\tau, \eta)$ pulse-by-pulse to $\tilde{s}(\tau, \eta)$ in the range-frequency domain, and $\tilde{s}(\tau, \eta)$ can be updated as

$$\begin{aligned} &\tilde{s}(\tau, \eta) \\ &= \mathcal{F}_d^{-1}\left[\mathcal{F}_d\left[\tilde{r}(\tau, \eta) + B_i \mathrm{sinc}(B_i \tau)\right] \times H_s(f_\tau, \eta)\right] \\ &= \tilde{r}(\tau - T_s(\eta), \eta) + B_i \mathrm{sinc}(B_i(\tau - T_s(\eta))). \end{aligned} \quad (18)$$

The Fourier transform is employed to focus PRFI cross-range, which can be expressed as

$$\begin{aligned} &\tilde{S}(\tau, f'_\eta) \\ &= \mathcal{F}_c\left[\tilde{r}(\tau - T_s(\eta), \eta) + B_i \mathrm{sinc}(B_i(\tau - T_s(\eta)))\right] \\ &= \tilde{R}\left(\tau - T_s(\eta), f'_\eta\right) \\ &\quad + B_i \mathrm{sinc}(B_i(\tau - T_s(\eta)))C_i \mathrm{sinc}\left(C_i f'_\eta\right) \end{aligned} \quad (19)$$

where C_i and f'_η denote the width of shifted PRFI along cross-range and frequency, respectively.

Within formula (19), the first term is the result of the SAR echo signal modulated by $H_I(f_\tau)$ and $H_s(f_\tau, \eta)$, whereas the second term is the LFM-PRFIs focused in both down-range and cross-range. The process by which LFI-PRFI focuses is shown schematically in Fig. 6.

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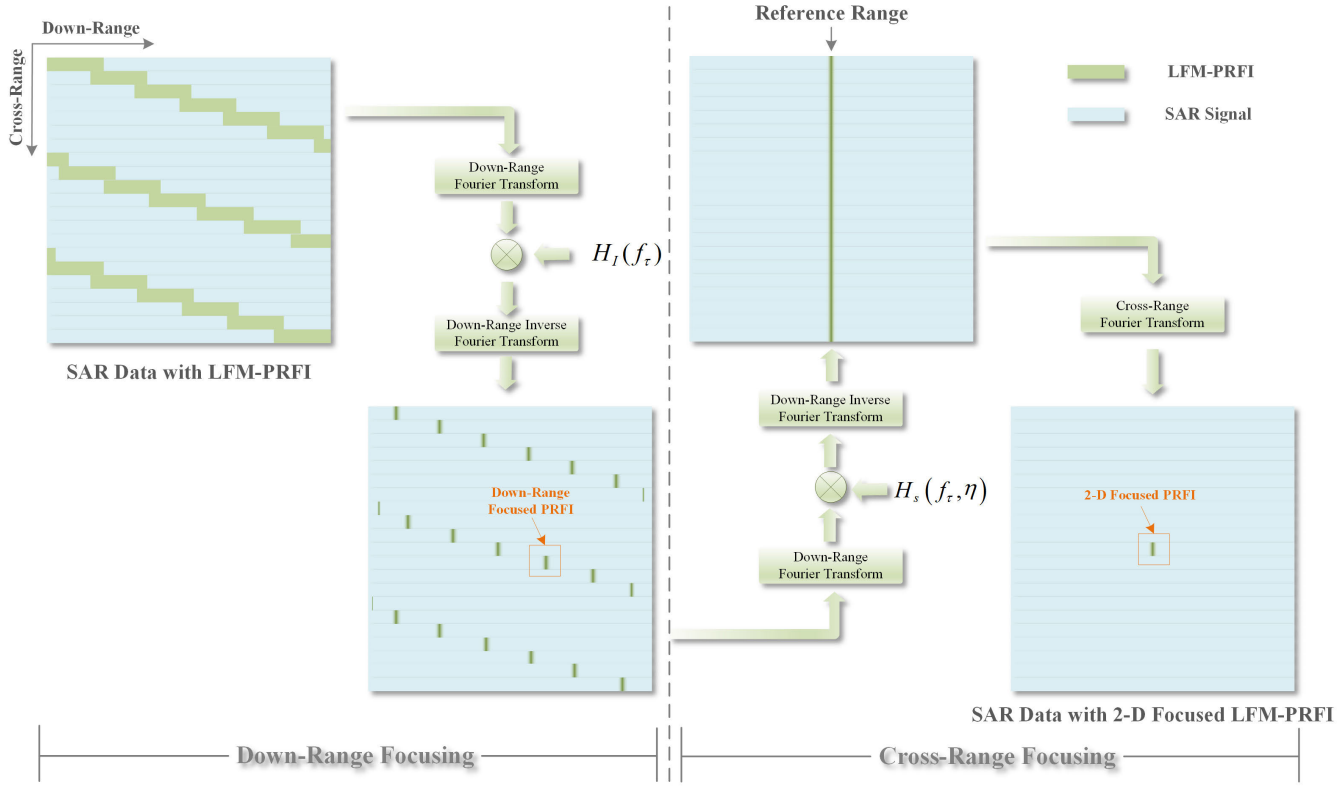


Fig. 6. Schematic of focusing LFM-PRFI.

B. LFM-PRFI Removal

After the above processing, the SAR signal can be regarded as noise compared to the focused PRFI. The proposed method employs mask processing to realize the function of filtering PRFI, and the function is designed to remove the focused PRFI, which can be expressed as

$$M(\tau, f_\eta') = \begin{cases} 1, & \left|\tilde{S}(\tau, f_\eta')\right| \le \xi \\ 0, & \text{else} \end{cases} \tag{20}$$

where ξ denotes the threshold used to distinguish PRFI from false alarms, and 0 and 1 mark the focused PRFI and false alarms, respectively. Then, the PRFI mitigation can be expressed as

$$\begin{aligned} S(\tau, f_\eta') &= \tilde{S}(\tau, f_\eta') \times M(\tau, f_\eta') \\ &= \{\tilde{R}(\tau - T_s(\eta), f_\eta') \\ &\quad + B_i \mathrm{sinc}(B_i(\tau - T_s(\eta)))C_i \mathrm{sinc}(C_i f_\eta')\} \times M(\tau, f_\eta') \\ &= \tilde{R}(\tau - T_s(\eta), f_\eta') \times M(\tau, f_\eta') \\ &\approx \tilde{R}(\tau - T_s(\eta), f_\eta'). \end{aligned} \tag{21}$$

It can be seen that the focused PRFIs, i.e., the sinc envelopes, are eliminated by mask processing. The time-domain notch filter removes the signal with length T_i, and the frequency-domain notch filter removes the spectrum with width B_i, whereas the useful SAR signal filtered by the proposed method is far less than the traditional notch filter-based methods. Therefore, the influence of masks on $\tilde{R}(\tau - T_s(\eta), f_\eta')$ can be ignored.

C. Data Restoration

Additional processing is required to restore $r(\tau, \eta)$ from $\tilde{R}(\tau - T_s(\eta), f_\eta')$ since the above processing introduces unwanted phase modulation to the SAR signal, and it can be expressed as

$$\begin{aligned} &s(\tau, \eta) \\ &= \mathcal{F}_d^{-1}\left[\mathcal{F}_d\left[\mathcal{F}_c^{-1}\left[S(\tau, f_\eta')\right]\right] \times H_s^*(f_\tau, \eta) \times H_I^*(f_\tau)\right] \\ &= \mathcal{F}_d^{-1}\left[\mathcal{F}_d\left[\tilde{r}(\tau - T_s(\eta), \eta)\right] \times H_s^*(f_\tau, \eta) \times H_I^*(f_\tau)\right] \\ &= r(\tau, \eta) \end{aligned} \tag{22}$$

where $*$ denotes the conjugate.

IV. Numerical Experiment

Extensive numerical experiments are carried out to test the mitigation performance of the proposed approach, and the experimental results are shown and analyzed in this section. Experiments based on simulated SAR data with different parameters of PRFI are employed to analyze the applicability of the proposed approach, and the root mean square error (RMSE) is employed to qualitatively evaluate the experimental results. Experiments based on the Sentinel-1 data from three LFM-PRFI-polluted cases are employed to further verify the effectiveness of the proposed approach, and the peak signal-to-noise ratio (PSNR) is employed to qualitatively evaluate the experimental results. Three representative methods, i.e., ESP [37], FNF [35], and TNF [33], are employed for comparison. In addition, information including simulation parameters,

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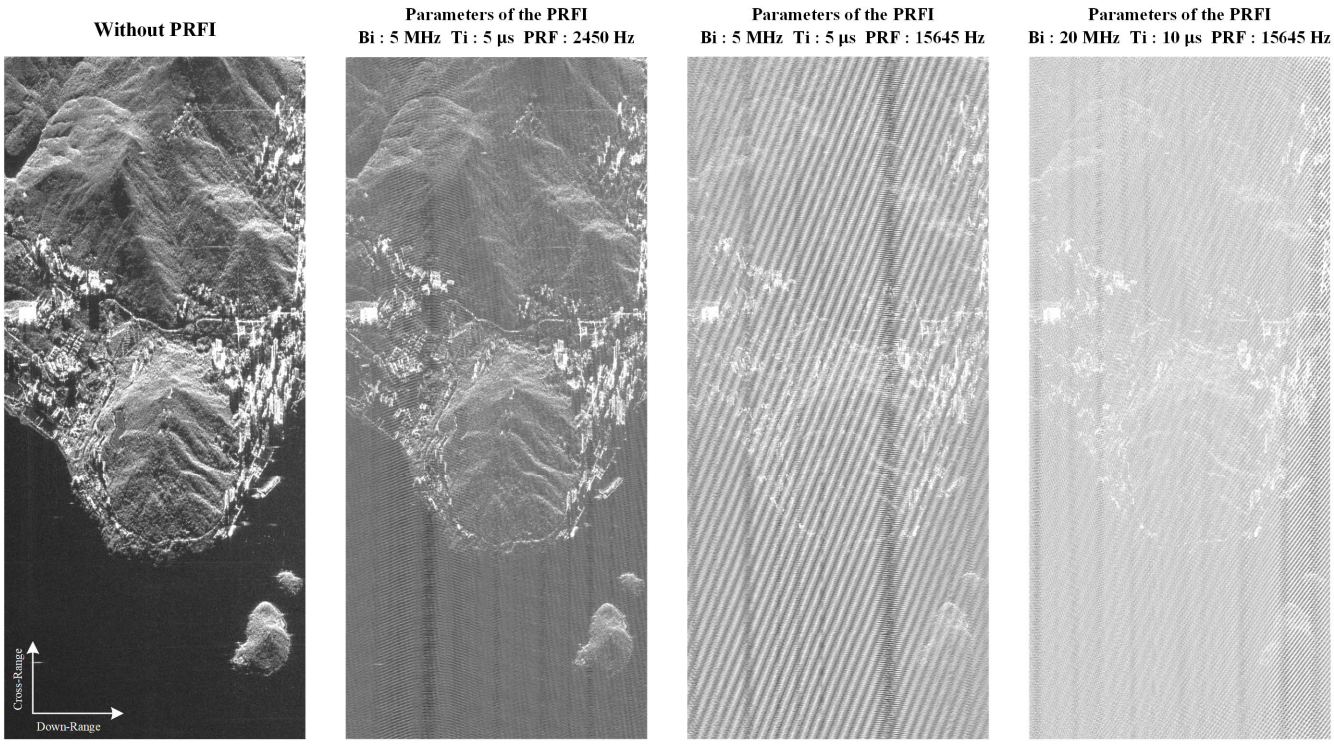


Fig. 7. Simulated SAR data and added with different LFM-PRFIs. (a) PRFI-free SAR image. (b)–(d) SAR image added LFM-PRFIs with different parameters. The parameters of the LFM-PRFI are marked at the top of the figures.

cases, data IDs, and qualitative indicators is given in detail in this section.

A. Evaluation Metrics

The first indicator introduced for the quantitative evaluation of experimental results is the RMSE. It reflects the deviation between the PRFI-free data and the PRFI-mitigated data, and the smaller the deviation, the better the mitigation performance of the method. Denoted the PRFI-free data and PRFI-mitigated data by $\mathbf{X}$ and $\tilde{\mathbf{X}}$, the RMSE can be calculated by

$$\mathrm{RMSE}\left(\mathbf{X}, \tilde{\mathbf{X}}\right) = \frac{\left\|\mathbf{X} - \tilde{\mathbf{X}}\right\|_F}{\|\mathbf{X}\|_F} \tag{23}$$

where $\|\mathbf{X}\|_F$ is the Frobenius norm of the matrix $\mathbf{X}$, which is defined as

$$\|\mathbf{X}\|_F = \sqrt{\sum_i \sum_j |x|^2}. \tag{24}$$

The second indicator introduced is the PSNR, which is used to evaluate SAR images after PRFI mitigation. It indicates the degree to which the image is distorted by interference and noise, and the larger the PSNR, the smaller the image distortion. The PSNR is defined as

$$\mathrm{PSNR}(\mathbf{X}_{\max}, \mathbf{X}_N) = 10\log_{10}\left(\frac{|\mathbf{X}_{\max}|^2 - \frac{1}{S_x S_y}\|\mathbf{X}_N\|_F}{\frac{1}{S_x S_y}\|\mathbf{X}_N\|_F}\right) \tag{25}$$

where $\mathbf{X}_{\max}$ and $\mathbf{X}_N$ denote the max value of the image and selected image to consider as noise, respectively. S_x and S_y are the sizes of the $\mathbf{X}_N$.

The third indicator introduced is the mean square deviation (MSD), which is used to measure the gray value fluctuation of the image. The MSD is defined as

$$\mathrm{MSD} = \frac{\left\|\tilde{\mathbf{X}} - \mathrm{mean}(\tilde{\mathbf{X}})\right\|_F}{N_a N_r}. \tag{26}$$

The larger the MSD of the image, the greater the fluctuation of the gray value of the image, and the richer the detail information of the image.

B. On the Mitigation Performance of the Simulated SAR Data

The simulation scene is selected in a port, where the ground objects and landforms are rich, and the simulation parameters are the same as in Table I. The LFM-PRFIs with different parameters are artificially added to the simulation scene, and the interferer is set in the center of the scene. Fig. 7 shows the simulated SAR data and added with different LFM-PRFIs, wherein Fig. 7(a) shows the PRFI-free SAR image and Fig. 7(b)–(d) shows the SAR image added LFM-PRFIs with different parameters, respectively. The RMSEs of the mitigation results are listed in Table II.

The parameters of the simulated PRFI are marked at the top of Fig. 7, and these parameters can be interpreted as follows. The bandwidth and pulsewidth of the simulated PRFI in Fig. 7(b) and (c) are consistent, but their PRF is different. The waveforms of PRFI in the two data are consistent, but the content of the latter is higher than that of the former (the content of the PRFI depends on its PRF), which is why the

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TABLE II
RMSEs OF FNF, ESP, TNF, AND PROPOSED APPROACH

PRFI \ Method		FNF	ESP	TNF	Proposed Approach
PRFI #1	$B_i = 5\ MHz \quad T_i = 5\ \mu s \quad PRF = 2450\ Hz$	0.5558	0.3467	0.2352	**0.2337**
PRFI #2	$B_i = 5\ MHz \quad T_i = 5\ \mu s \quad PRF = 15645\ Hz$	0.5731	0.4185	0.3441	**0.3294**
PRFI #3	$B_i = 20\ MHz \quad T_i = 10\ \mu s \quad PRF = 15645\ Hz$	0.5895	0.5971	0.4396	**0.4368**

PRFI artifacts are more prominent in the latter image. Furthermore, the PRFs of the simulated PRFI in Fig. 7(c) and (d) are consistent, but the bandwidth and pulsewidth are different, which means that, in the case of the same PRF, the larger the bandwidth and pulsewidth of the PRFI, the more serious their impact on the SAR image. In addition, it should be pointed out that, in the experiment, the simulated SAR data were normalized, and the SINR of the simulated PRFI is set to -3 dB.

The mitigation results of the FNF, ESP, TNF, and the proposed approach are shown in Fig. 8, respectively, in which from top to bottom are the experimental results of adding LFM-PRFIs with different parameters, and from left to right are the experimental results of the above four methods; three representative ROIs, i.e., mountains, ports, and buildings, are boxed in the figure and enlarged on the right. Within the first row of Fig. 8, the mitigation performance of the four methods differs little when the bandwidth and pulsewidth of PRFI are small. However, as can be seen from the enlarged ROI on the right, the FNF method loses image resolution due to the introduction of spectral gaps, and the ESP method leaves a part of PRFI artifacts. Within the second row of Fig. 8, the difference in the mitigation performance of the four methods increases with the increase in the PRF of PRFI. The higher the PRFI, the more likely the SAR receiver will receive the PRF, and the greater the PRFI content in the data; in this case, the drawbacks of time-domain-based mitigation methods begin to show. As can be seen in the ship on the right, due to the high PRFI content of the data, the ESP method excessively penalizes the eigenvalues, resulting in missing principal components of the SAR signal, thus introducing azimuth ambiguity. The SNR of the data in the results of the FNF and TNF methods is significantly lower; besides this, the FNF method deteriorates the resolution of the ship. An interesting phenomenon can be seen in the third row of Fig. 8. With the further deterioration of the PRFI parameters, the mitigation performance of the traditional methods drops significantly. As analyzed above, the performance of the FNF method deteriorates with the increase in the PRFI bandwidth, while the performance of the TNF method deteriorates with the increase in the PRFI pulsewidth. These phenomena were demonstrated in the third set of simulation experiments. As can be seen from the buildings marked by blue boxes, the mitigation performance of the proposed approach is better than that of the other methods. Further quantitative analysis shows that the RMSE of the proposed approach is the smallest in the three groups of experiments, which means that the difference between the mitigated data and the pure data is small, thus indicating that the mitigation performance of the proposed method is better than other methods.

C. On the Mitigation Performance of the Real SAR Data

1) Data Description: Three example cases of LFM-PRFI contaminated SAR data, i.e., inland scene, near-shore scene, and far coast scene, acquired from ESA Sentinel-1 are employed to further verify the effectiveness of the proposed approach. The descriptions of the dataset are shown in Figs. 4 and 9, where data IDs are marked below these figures. The detailed data information is expressed in Table III for the reference of interested readers.

The first example case is introduced and analyzed in Section II. The second example case is a near-shore area located in Copenhagen, Denmark, which is the largest city and port in this country. Fig. 9(a)–(d) shows the geolocation, amplitude image of the raw data, amplitude image of the range spectrum, and amplitude image of the experimental data, respectively. In this case, the bandwidth of the LFM-PRFI is higher than half of the SAR signal bandwidth, and interference-introduced artifacts are overlaid on the image, obscuring the city and port area. The third example case is a far coast area located in the English Channel. Fig. 9(e)–(h) shows the geolocation, amplitude image of the raw data, amplitude image of the range spectrum, and amplitude image of the experimental data, respectively. Since this area is sea clutter [55] accompanied by sparse ships, the energy of the SAR signal is weak, and only the frequency components of the PRFI can be seen on the range spectrum, whereas the artifacts caused by PRFI are dominant in the SAR image, and no man-made targets, such as ships, can be seen.

2) Experimental Results for the First Example Case: Mitigation results of the first example case are shown in Fig. 10. The significant artifacts of PRFI seriously covered the observation scene, which is shown in Fig. 4(d). Effective scene information, i.e., urban topography, is recovered after performing mitigation processing. According to the previous analysis, the bandwidth of PRFI is 14.43 MHz, while the bandwidth of the SAR signal is only 42.79 MHz, which means that the FNF method will generate a large number of spectrum gaps, resulting in abnormal target sidelobes and SNR deterioration in the final SAR image. As can be seen from the ROI on the right, details such as towns and rivers have been eliminated, which is intolerable for some SAR application processing. Due to bandwidth limitations, the ESP method fails to mitigate all PRFI, and as can be seen from the ROI on the right, residual PRFI still looms over the town. In contrast, the TNF method

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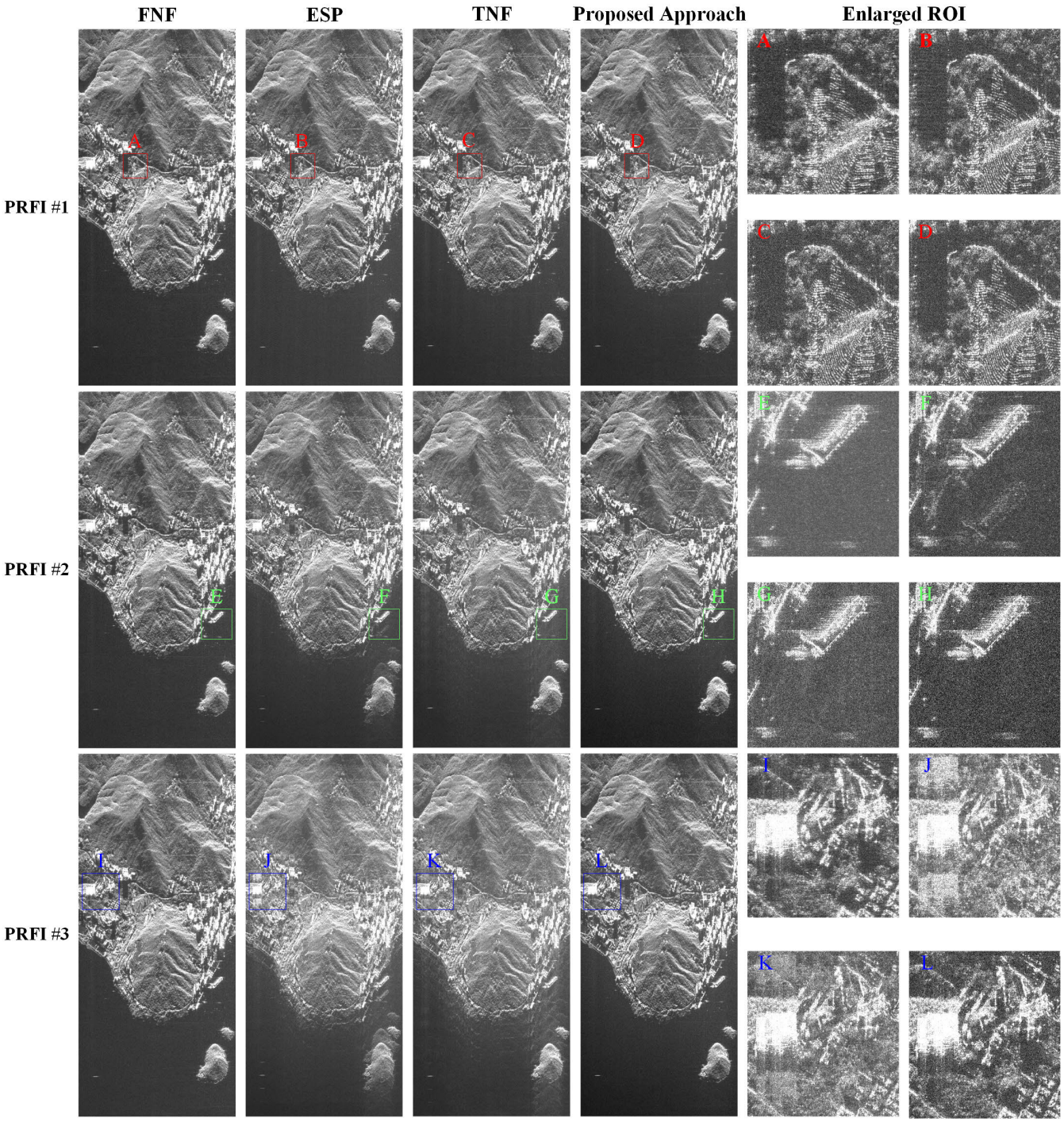


Fig. 8. Mitigation results of the FNF, ESP, TNF, and the proposed approach, in which from (top) to (bottom) are the experimental results of adding LFM-PRFIs with different parameters, and from (left) to (right) are the experimental results of the above four methods; three representative ROIs, i.e., mountains, ports, and buildings, are boxed in the figure and enlarged on (right).

is superior to the FNF and ESP methods, and the PRFIs are eliminated while retaining the details of the image. From a qualitative point of view, the image quality in the mitigation results of the proposed method is significantly better than that of other methods. It should be noted that the river area marked by the red box in the second column will be used to qualitatively analyze the PSNR of the image.

3) Experimental Results for the Second Example Case: Mitigation results of the second example case are shown in Fig. 11, where the powerful PRFI artifacts are mitigated, and scene information is restored. The bandwidth of PRFI in this data covers almost the entire SAR spectrum, as shown in Fig. 9(c), severely limiting the mitigation performance of the FNF and ESP methods. From the first and second rows of Fig. 11, it can be seen that the residual PRFI is still visible in some low SNR areas, such as the sea surface. As can be seen in the area marked by the red circle, the recovered scene is blurred, which means that a large number of useful

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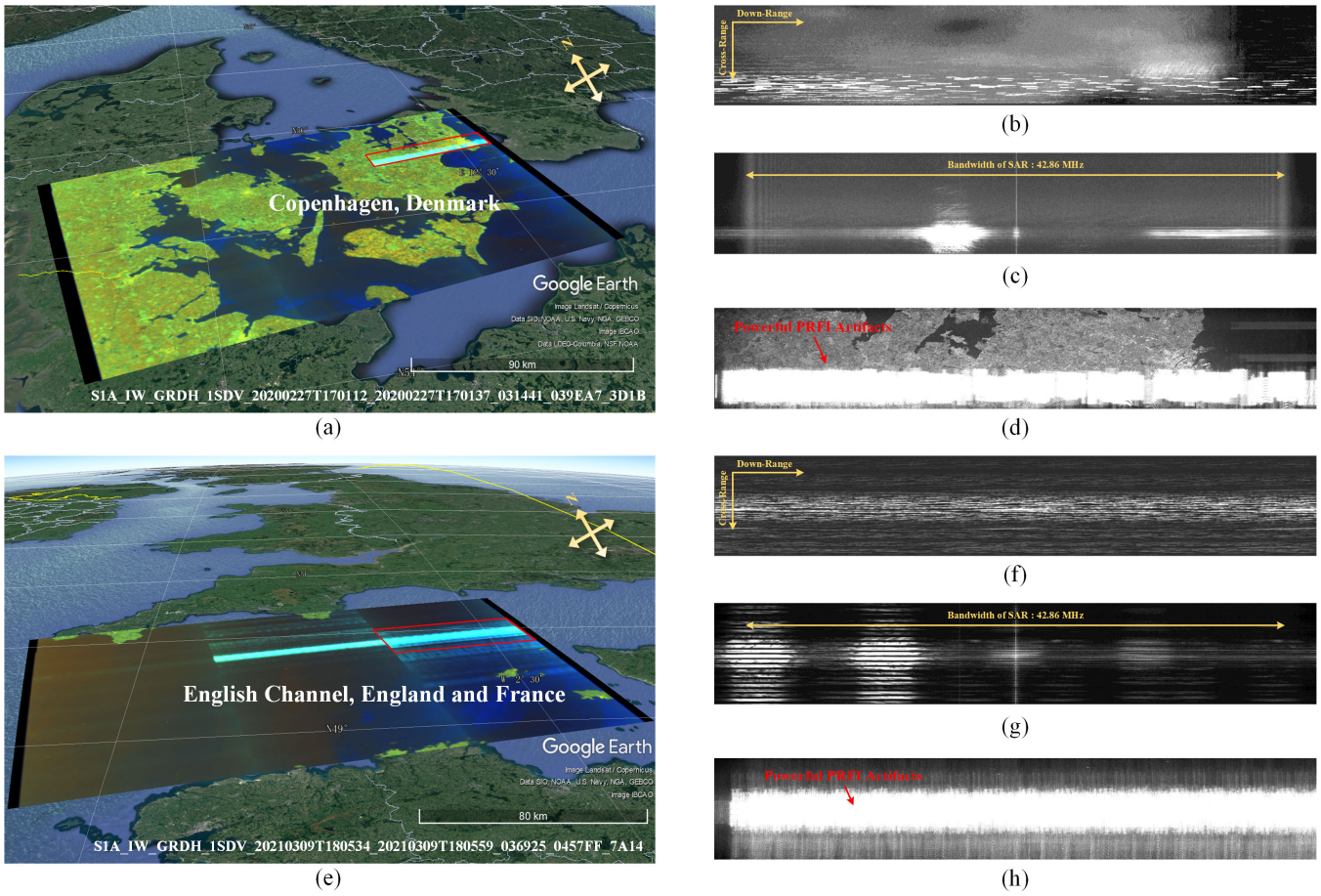


Fig. 9. Descriptions of the dataset. (a) and (e) Geolocations of the experimental data. (b) and (f) Amplitude image of the raw data. (c) and (g) Amplitude image of the range spectrum. (d) and (h) Amplitude image of the experimental data.

TABLE III
DATA INFORMATION

SAR Image	Fig. 4 (a)	Fig. 9 (a)	Fig. 9 (e)
Sensor	Sentinel-1A	Sentinel-1A	Sentinel-1A
Band	C	C	C
Mode	IW	IW	IW
Polarization	VH	VH	VH
Acquisition Date	28 Mar. 2019	27 Feb. 2020	9 Mar. 2021
Interfered Area	Anhui, China	Copenhagen, Denmark	English Channel, England and France
Location	E 115.43, N 33.51	W 3.28, N 49.54	E 12.24, N 55.41
Swath	The 3rd Swath	The 3rd Swath	The 3rd Swath
Burst	The 4th Burst	The 9th Burst	The 8th Burst

SAR signals are missed while mitigating PRFI. In contrast, the mitigation performance of the TNF method and the proposed approach is better than that of the FNF and ESP methods because they are not limited to the bandwidth of PRFI. As can be seen in the area marked by the red circle in the third and fourth rows, the mitigation results of the proposed method show the topography information more clearly. As can be seen on the right of Fig. 11, the building outline of the proposed approach is clearer, and the clutter level is lower, which further illustrates its effectiveness.

4) Experimental Results for the Third Example Case: Mitigation results of the third example case are shown in Fig. 12. This scene is the sea surface and accompanied by sparse ships. The coherent accumulated energy of the SAR signal is much lower than that of PRFI, which is why the scene is full of interference and the target cannot be seen. Such a harsh scene is filled with a lot of PRFI and is undoubtedly challenging for PRFI mitigation. In the results of the FNF method, the sidelobes of the ships are abnormal both in down-range and cross-range, and the same phenomenon is seen in the results of the ESP method, where the residual PRFI still affects the interpretation of the ships. In the results of the TNF method, the sidelobes of the ship in cross-range are more obvious. The contours of a marked ship target are shown on the right-hand side of Fig. 11. The main energy of the ship is concentrated above -20 dB, whereas, in the results

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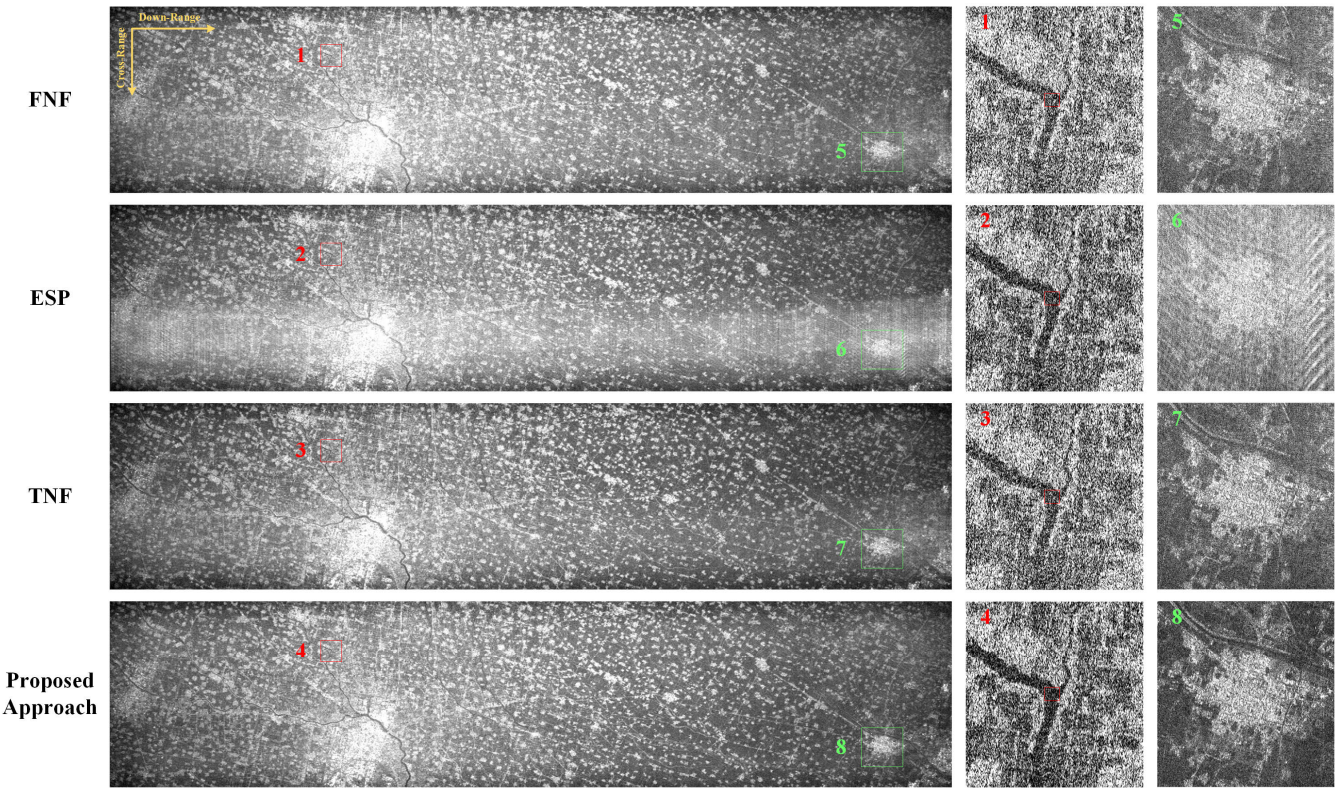


Fig. 10. Mitigation results of the first example case by the FNF, ESP, TNF, and proposed approach. Images of the first column are the mitigation results. Images of the second and third columns are the ROIs.

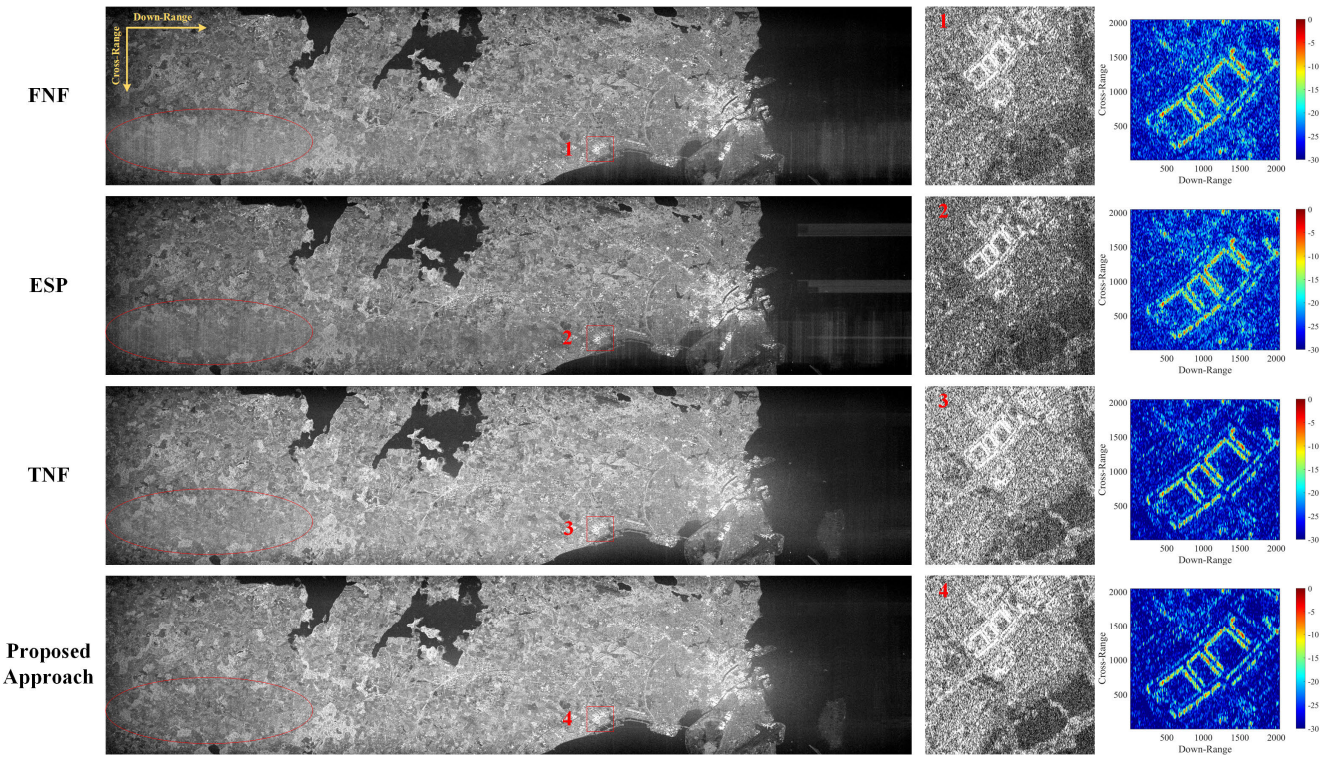


Fig. 11. Mitigation results of the second example case by the FNF, ESP, TNF, and proposed approach. Images of the first column are the mitigation results. Images of the second column are the ROIs. Images of the third column are the enlarged building area in ROIs.

of FNF and TNF, some clutter of −30 dB is more obvious, and the range sidelobe of the ESP method is significantly high. In contrast, the ship outline of the proposed method is clear, and the level of the sidelobe is acceptable, indicating that its mitigation performance is better than that of the other methods.

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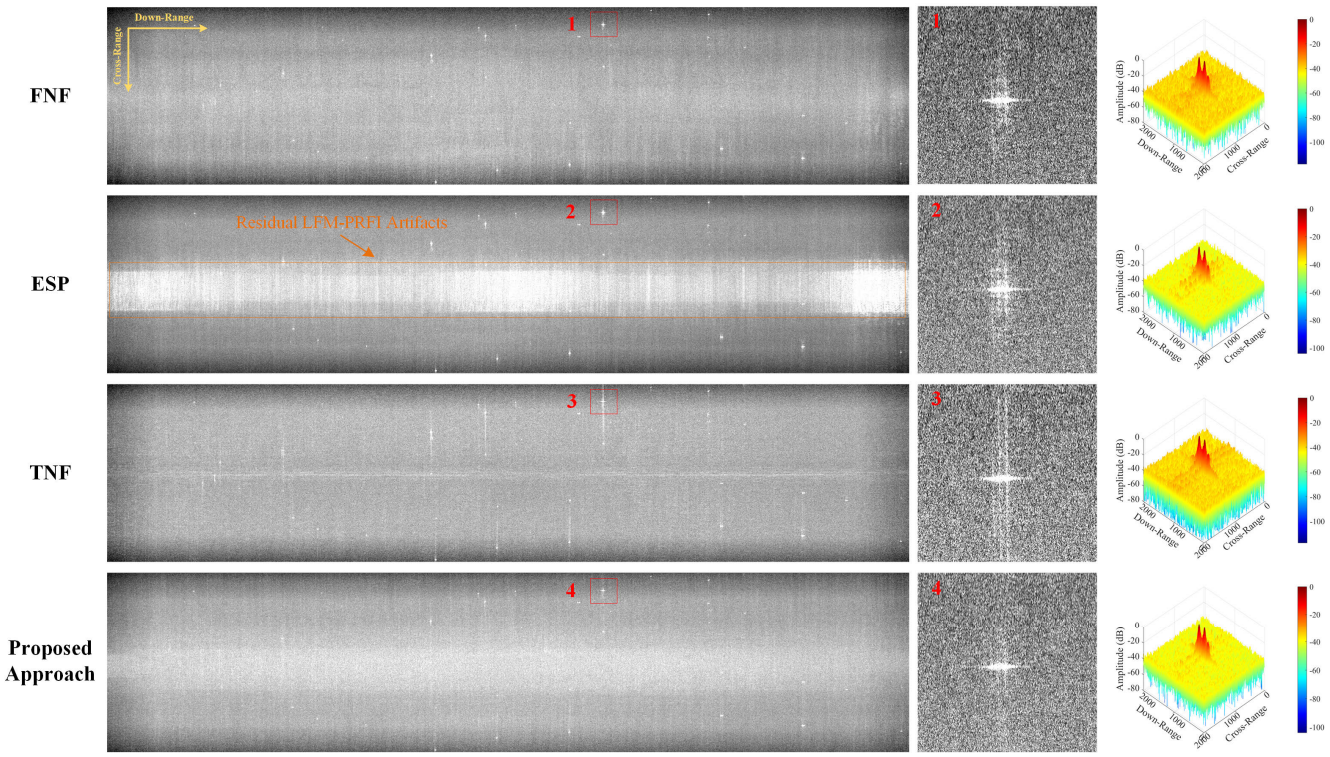


Fig. 12. Mitigation results of the third example case by the FNF, ESP, TNF, and proposed approach. Images of the first column are the mitigation results. Images of the second column are the ROIs. Images of the third column are the contours of the ship targets marked with red boxes.

TABLE IV
EVALUATION METRICS OF FOUR METHODS

Example Case \ Method	PSNR				MSD			
	FNF	ESP	TNF	Proposed Approach	FNF	ESP	TNF	Proposed Approach
Example Case #1	44.02 dB	44.31 dB	44.81 dB	**45.37 dB**	0.0370	0.0379	0.0373	**0.0378**
Example Case #2	61.12 dB	62.69 dB	63.17 dB	**63.41 dB**	0.0115	0.0112	0.0166	**0.0167**
Example Case #3	44.17 dB	44.08 dB	43.89 dB	**45.91 dB**	0.0228	**0.0241**	0.0187	0.0231

The PSNR and MSD of the mitigation results are further calculated, which are listed in Table IV. The PSNR of the proposed approach is superior to other methods, indicating that the proposal has a low impact on SAR data, and the image details are well preserved, which further illustrates its effectiveness and superiority. It should be pointed out that, for the third example case, the higher MSD of the ESP method than the other methods is due to the residual PRFI artifacts in the image. On the whole, the MSD of the proposed method is better than other methods, indicating that the details of the image are rich and the image is restored to the ideal level.

D. On the Discussion of the Computational Complexity

Within the FNF method and the proposed method, the Fourier transform needs to implement in discrete cases, and therefore, the fast version of the Fourier transform, i.e., fast Fourier transform (FFT), is employed in the experiment. Denoting the sampling number by $\mathcal{N}$ and its exact power as 2, at least $\mathcal{O}(\mathcal{N}\log_2\mathcal{N})$ flops are required for performing FFT (the same timing cost for its inverse version).

Within the FNF method, an FFT and its inverse version are required, and the total arithmetic complexity is

$$\mathcal{C}_{\text{FNF}} = 2\mathcal{O}\left(N_a N_r \log_2 N_r\right) + \mathcal{O}(N_a N_r). \tag{27}$$

The total arithmetic complexity of the ESP method [14] is

$$\mathcal{C}_{\text{ESP}} = \mathcal{O}\left(N_a\left(L^3 + N_r + 1 - L\right)\right) \tag{28}$$

where L is the dimension of subspace.

Within the TNF method, the main timing cost comes from MIAA-based missing data recovery. As introduced in [34], at least $\mathcal{O}\left(N_g^3 + N_a\log_2 N_a\right)$ flops are required to recover the missed data, and the total arithmetic complexity is

$$\mathcal{C}_{\text{TNF}} = \mathcal{O}\left(N_r\left(N_g^3 + N_a\log_2 N_a\right)\right) \tag{29}$$

where N_g denotes the number of gaps.

As far as the proposed method is concerned, the main timing cost comes from PRFI-matched filtering, shift, and cross-range focusing, which requires the quartic FFT operation and the quartic inverse version. Then, the total arithmetic complexity

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of the proposed method is

$$\mathcal{C}_{\mathrm{PA}} = 3\mathcal{O}\left(N_a N_r \log_2 N_r\right) + \mathcal{O}\left(N_r N_a \log_2 N_a\right). \quad (30)$$

It can be easily found through the above analysis that the ESP and TNF methods are time-consuming. Although the computational complexity of the FNF method is better than that of the proposed method, its powerlessness for wideband PRFI is unacceptable. Taken together, the mitigation performance of the proposed method is better than the other three methods.

V. Conclusion

The problem caused by a wideband version of PRFI, i.e., LFM-PRFI originating from the ground- and space-based radar sensors, is increasingly common in SAR systems and cannot be ignored. This article introduces the model, mechanism, and characteristics of LFM-PRFI and characterizes them by multiple real cases acquired from ESA Sentinel-1. A joint down-range and cross-range filtering approach to mitigate LFM-PRFI is proposed. The proposal focuses on the energy of LFM-PRFI in two orthogonal directions and then filters it out through the designed filter to achieve the purpose of mitigating LFM-PRFI. Compared with the traditional notch filter-based methods, the proposal is easy to implement, time-saving, and efficient, and it overcomes the introduction of abnormal target responses and ghosts by the traditional methods.

Experiments based on simulated SAR data interfered with by different parameters of PRFI and real spaceborne SAR data are performed. Specifically, simulation results show that the mitigation performance of the proposed approach is better than that of the traditional methods, and it is suitable for mitigating LFM-PRFI with large bandwidth and pulsewidth. Experimental results based on ESA Sentinel-1 data show that the proposed approach is effective, and the mitigation performance is better than the traditional notch filter-based methods. Detailed information about the used Sentinel-1 data is listed in this article for the reference of interested readers.

It needs to be pointed out that the shortcomings of the proposed method are given as follows: 1) the proposed method is only suitable for mitigating LFM PRFI, and its mitigation performance depends on the accuracy of FM rate estimation and 2) if the PRFI in SAR data comes from multiple interference sources, then the iterative processing is required. Three studies will be further carried out in the future.

1) Apply the proposed approach to more SAR datasets.

2) Expand the proposed approach to mitigate more complicated types of PRFI.

3) Improve the mitigation performance further based on a priori knowledge, i.e., parameters of the PRFI.

The authors hope that the proposal can be further validated on more spaceborne SAR platforms and serve as a feasible solution to the LFM-PRFI problem.

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References

[1] A. Reigber et al., "Very-high-resolution airborne synthetic aperture radar imaging: Signal processing and applications," *Proc. IEEE*, vol. 101, no. 3, pp. 759–783, Mar. 2013.

[2] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, and I. P. K. Hajnsek, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, May 2013.

[3] G. Krieger et al., "TanDEM-X: A satellite formation for high-resolution SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3317–3341, Nov. 2007.

[4] P. Huang, H. Yang, X.-G. Xia, Z. Zou, X. Liu, and G. Liao, "A novel sea clutter rejection algorithm for spaceborne multichannel radar systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5117422.

[5] Y. Zhang et al., "First demonstration of multipath effects on phase synchronization scheme for LT-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2590–2604, Apr. 2020.

[6] S. Niu, D. Zhu, G. Jin, Y. Cheng, and Y. Wang, "A novel transmitter-interpulse phase coding MIMO-radar for range ambiguity separation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5238216.

[7] S. Misra and P. de Matthaeis, "Passive remote sensing and radio frequency interference (RFI): An overview of spectrum allocations and RFI management algorithms," *IEEE Geosci. Remote Sens. Mag.*, vol. 2, no. 2, pp. 68–73, Jun. 2014.

[8] M. Spencer and F. Ulaby, "Spectrum issues faced by active remote sensing: Radio frequency interference and operational restrictions technical committees," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 1, pp. 40–45, Mar. 2016.

[9] M. Shimada, O. Isoguchi, T. Tadono, and K. Isono, "PALSAR radiometric and geometric calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 3915–3932, Dec. 2009.

[10] S-1 Mission Performance Center (S-1 MPC). (2016). *S1-A Annual Performance Report for 2015*. [Online]. Available: https://sentinel.esa.int/documents/247904/0/Sentinel-1A-Annual-Performance-Report-2015/1285c804-d83e-48e9-a934-4db30208f28a

[11] S-1 Mission Performance Center (S-1 MPC). (2018). *Sentinel-1 Long Duration Mutual Interference*. [Online]. Available: https://sentinel.esa.int/documents/247904/2142675/Sentinel-1-Long-Duration-Mutual-Interference

[12] S-1 Mission Performance Center (S-1 MPC). (2022). *SAR-MPC Sentinel-1: Using the RFI Annotations*. [Online]. Available: https://sentinels.copernicus.eu/documents/247904/1653442/DI-MP COTH-0540-1-0-RFI-Tech-Note.pdf

[13] Z. Lv, N. Li, Z. Guo, and J. Zhao, "Detection and mitigation of mutual RFI in C-band SAR: A case study of Chinese GaoFen-3," in *Proc. IEEE Radar Conf.*, May 2021, pp. 1–5.

[14] N. Li, Z. Lv, and Z. Guo, "Observation and mitigation of mutual RFI between SAR satellites: A case study between Chinese GaoFen-3 and European Sentinel-1A," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5112819.

[15] H. Yang, M. Tao, S. Chen, F. Xi, and Z. Liu, "On the mutual interference between spaceborne SARs: Modeling, characterization, and mitigation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8470–8485, Oct. 2021.

[16] 2022 S-1 RFI Maps. (2022). *Aresys Earth Observation*. [Online]. Available: https://s1rfimap.aresys.it

[17] Z. Liu, G. Liao, and Z. Yang, "Time variant RFI suppression for SAR using iterative adaptive approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1424–1428, Nov. 2013.

[18] H. Liu, D. Li, Y. Zhou, and T.-K. Truong, "Joint wideband interference suppression and SAR signal recovery based on sparse representations," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1542–1546, Sep. 2017.

[19] L. H. Nguyen, T. Tran, and T. Do, "Sparse models and sparse recovery for ultra-wideband SAR applications," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 2, pp. 940–958, Apr. 2014.

[20] Y. Huang, G. Liao, Z. Zhang, Y. Xiang, J. Li, and A. Nehorai, "Fast narrowband RFI suppression algorithms for SAR systems via matrix-factorization techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 250–262, Jan. 2019.

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[21] J. Su, H. Tao, M. Tao, L. Wang, and J. Xie, "Narrow-band interference suppression via RPCA-based signal separation in time-frequency domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 5016–5025, Nov. 2017.

[22] J. Shen, B. Han, Z. Pan, G. Li, Y. Hu, and C. Ding, "Learning time–frequency information with prior for SAR radio frequency interference suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5239716.

[23] M. Tao, J. Su, Y. Huang, and L. Wang, "Mitigation of radio frequency interference in synthetic aperture radar data: Current status and future trends," *Remote Sens.*, vol. 11, no. 20, p. 2438, Oct. 2019.

[24] T. R. Miller, J. W. McCorkle, and L. C. Potter, "Near-least-squares radio frequency interference suppression," in *Proc. SPIE*, vol. 2487, pp. 72–83, Jun. 1995.

[25] B. Ulug, "An algorithm for sinusoidal interference reduction using iterative maximum likelihood estimation techniques," M.S. thesis, Dept. Elect. Eng., Ohio State Univ., Columbus, OH, USA, Mar. 1992.

[26] X. Luo, L. M. H. Ulander, J. Askne, G. Smith, and P. O. Frolind, "RFI suppression in ultra-wideband SAR systems using LMS filters in frequency domain," *Electron. Lett.*, vol. 37, no. 4, pp. 241–243, Feb. 2001.

[27] M. Braunstein, J. M. Ralston, and D. A. Sparrow, "Signal processing approaches to radio frequency interference (RFI) suppression," *Proc. SPIE*, vol. 2230, pp. 190–208, Jun. 1994.

[28] S. Buckreuss and R. Horn, "E-SAR P-band SAR subsystem design and RF-interference suppression," in *Proc. IGARSS*, Jul. 1998, pp. 466–468.

[29] T. Miller, L. Potter, and J. McCorkle, "RFI suppression for ultra-wideband radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 4, pp. 1142–1156, Oct. 1997.

[30] A. Potsis et al., "P-band interference study using information from the new E-SAR listen only channel mode of operation," in *Proc. EUSAR*, 2000, pp. 415–418.

[31] W. Han, X. Bai, W. Fan, L. Wang, and F. Zhou, "Wideband interference suppression for SAR via instantaneous frequency estimation and regularized time-frequency filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5208612.

[32] W. Han and F. Zhou, "SAR wideband interference suppression method using second-order multisynchrosqueezing transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5230215.

[33] N. Li, Z. Lv, Z. Guo, and J. Zhao, "Time-domain notch filtering method for pulse RFI mitigation in synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[34] N. Li, Z. Lv, and Z. Guo, "Pulse RFI mitigation in synthetic aperture radar data via a three-step approach: Location, notch, and recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5225617.

[35] G. Cazzaniga and A. M. Guarnieri, "Removing RF interferences from P-band airplane SAR data," in *Proc. IGARSS*, Jun. 1996, pp. 1845–1847.

[36] W. Xu, W. Xing, C. Fang, P. Huang, and W. Tan, "RFI suppression based on linear prediction in synthetic aperture radar data," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 12, pp. 2127–2131, Dec. 2021.

[37] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Eigensubspace-based filtering with application in narrow-band interference suppression for SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 75–79, Jan. 2007.

[38] F. Zhou, M. Tao, X. Bai, and J. Liu, "Narrow-band interference suppression for SAR based on independent component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 4952–4960, Oct. 2013.

[39] F. Zhou, M. Xing, X. Bai, G. Sun, and Z. Bao, "Narrow-band interference suppression for SAR based on complex empirical mode decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 423–427, Jul. 2009.

[40] M. Tao, F. Zhou, J. Liu, Z. Zhang, and Z. Bao, "Narrow-band interference mitigation for SAR using independent subspace analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5289–5301, Sep. 2014.

[41] J. Feng, H. Zheng, Y. Deng, and D. Gao, "Application of sub-band spectral cancellation for SAR narrow-band interference suppression," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 190–193, Mar. 2012.

[42] B. Chen, Z. Lv, P. Lu, G. Shu, Y. Huang, and N. Li, "Extension and evaluation of SSC for removing wideband RFI in SLC SAR images," *Remote Sens.*, vol. 14, no. 17, p. 4294, Aug. 2022.

[43] H. Yang, K. Li, J. Li, Y. Du, and J. Yang, "BSF: Block subspace filter for removing narrowband and wideband radio interference artifacts in single-look complex SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5211916.

[44] H. Yang, Y. He, Y. Du, T. Zhang, J. Yin, and J. Yang, "Two-dimensional spectral analysis filter for removal of LFM radar interference in spaceborne SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219016.

[45] Y. Huang et al., "HRWS SAR narrowband interference mitigation using low-rank recovery and image-domain sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5217914.

[46] L. H. Nguyen and T. D. Tran, "Efficient and robust RFI extraction via sparse recovery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2104–2117, Jun. 2016.

[47] S. Joy, L. H. Nguyen, and T. D. Tran, "Joint down-range and cross-range RFI suppression in ultra-wideband SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3136–3149, Apr. 2021.

[48] Y. Huang, G. Liao, J. Xu, and J. Li, "Narrowband RFI suppression for SAR system via efficient parameter-free decomposition algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3311–3322, Jun. 2018.

[49] Y. Huang, G. Liao, Y. Xiang, Z. Zhang, J. Li, and A. Nehorai, "Reweighted nuclear norm and reweighted Frobenius norm minimizations for narrowband RFI suppression on SAR system," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5949–5962, Aug. 2019.

[50] Y. Huang, G. Liao, L. Zhang, Y. Xiang, J. Li, and A. Nehorai, "Efficient narrowband RFI mitigation algorithms for SAR systems with reweighted tensor structures," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9396–9409, Nov. 2019.

[51] Y. Huang, L. Zhang, J. Li, Z. Chen, and X. Yang, "Reweighted tensor factorization method for SAR narrowband and wideband interference mitigation using smoothing multiview tensor model," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3298–3313, May 2020.

[52] Y. Huang et al., "An efficient graph-based algorithm for time-varying narrowband interference suppression on SAR system," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8418–8432, Oct. 2021.

[53] H. Zhang, L. Min, J. Lu, J. Chang, Z. Guo, and N. Li, "An improved RFI mitigation approach for SAR based on low-rank sparse decomposition: From the perspective of useful signal protection," *Remote Sens.*, vol. 14, no. 14, p. 3278, Jul. 2022.

[54] Z. Tang, Y. Deng, and H. Zheng, "RFI suppression for SAR via a dictionary-based nonconvex low-rank minimization framework and its adaptive implementation," *Remote Sens.*, vol. 14, no. 3, p. 678, Jan. 2022.

[55] P. Huang, Z. Zou, X.-G. Xia, X. Liu, G. Liao, and Z. Xin, "Multichannel sea clutter modeling for spaceborne early warning radar and clutter suppression performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8349–8366, Oct. 2021.



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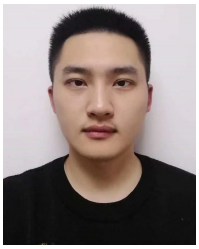
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